

CHAPTER - 6 EQUILIBRIUM

Key Idea

At equilibrium, no change in macroscopic properties.

“Equilibrium is dynamic, not static.”

1. EQUILIBRIUM

The state in a reversible reaction in which the **rates** of the **forward** and **reverse** reactions become equal, so that there is **no net change** in the concentrations of reactants and products.

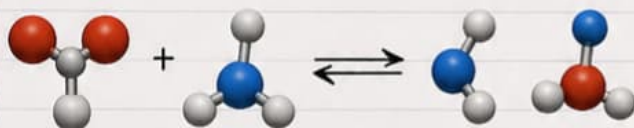
In reversible reaction:



k_f = rate constant of forward reaction

k_r = rate constant of reverse reaction

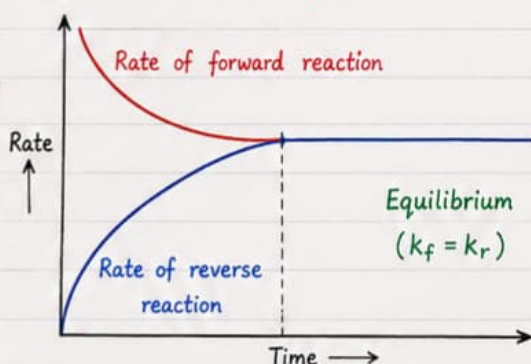
3. REVERSIBLE REACTION (3D VIEW)



(Example: $N_2(g) + 3H_2(g) \rightleftharpoons 2NH_3(g)$)

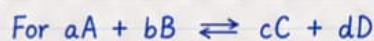
2. DYNAMIC NATURE OF EQUILIBRIUM

At equilibrium both forward and reverse reactions continue at **equal rates**.



4. EQUILIBRIUM CONSTANT (K_c)

For a balanced reversible reaction at a given temperature, the **equilibrium constant** is the ratio of the product of equilibrium concentrations of products to that of reactants.



$$K_c = \frac{[C]^c [D]^d}{[A]^a [B]^b}$$

K_c has **no units**.

5. FACTORS AFFECTING EQUILIBRIUM



① **Concentration**: Increasing concentration of reactants increases rate of forward reaction.



② **Pressure**: Increasing pressure (or decreasing volume) favours the side with less moles of gas.



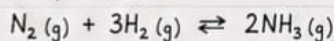
③ **Temperature**: Increasing temperature favours the endothermic reaction.



④ **Catalyst**: It does not change the position of equilibrium but helps in attaining it quickly.

6. EXAMPLES

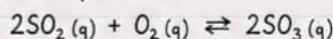
① Haber process:



Exothermic reaction



② Contact process:



Exothermic reaction



KEY TAKEAWAY

- Equilibrium is **dynamic** state in reversible reactions.
- At equilibrium, **rate forward** = **rate reverse**
- Equilibrium constant (K_c) depends only on temperature.
- Factors like concentration, pressure, temperature and catalysts affect equilibrium.

STAY
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CHAPTER - 6 : EQUILIBRIUM

At equilibrium,
forward = reverse
(microscopically)

7. LAW OF CHEMICAL EQUILIBRIUM

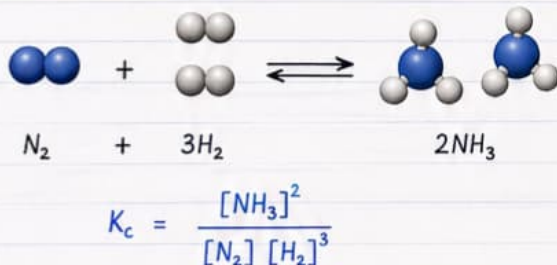
At a given temperature, for a reversible reaction, the ratio of the product of equilibrium concentrations of products to that of reactants, each raised to their stoichiometric coefficients, is **constant**.

For $aA + bB \rightleftharpoons cC + dD$

$$K_c = \frac{[C]^c [D]^d}{[A]^a [B]^b}$$

★ K_c is same for a reaction at a given temperature and **does not depend** on initial concentrations.

Example 1 : For $N_2(g) + 3H_2(g) \rightleftharpoons 2NH_3(g)$

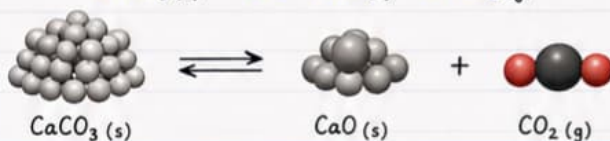
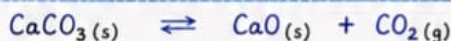


K_c depends only on temperature.
If temperature is constant,
 K_c remains constant.

8. HETEROGENEOUS EQUILIBRIUM

When a reversible reaction involves more than one phase, the concentration of pure solids and liquids remain constant and are **not** included in the equilibrium expression.

Example 2 :



Equilibrium expression :

$$K_c = [CO_2]$$

★ Solids and pure liquids have constant concentration = 1, so they are omitted.

9. RELATION BETWEEN K_c AND K_p

For a gaseous equilibrium reaction,
 K_p and K_c are related as :

$$K_p = K_c (RT)^{\Delta n_g}$$

where,

R = gas constant

T = absolute temperature

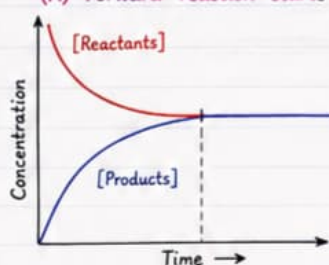
Δn_g = (moles of gaseous products) -
(moles of gaseous reactants)

Special cases :

- If $\Delta n_g = 0$, then $K_p = K_c$
- If $\Delta n_g > 0$, then $K_p > K_c$
- If $\Delta n_g < 0$, then $K_p < K_c$

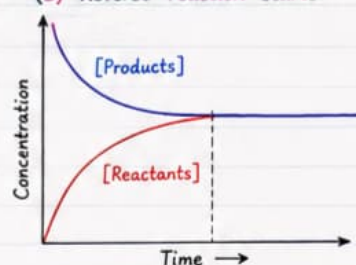
10. EQUILIBRIUM GRAPH (CONCENTRATION vs TIME)

(A) Forward reaction starts



At equilibrium,
rate forward = rate reverse
concentrations remain constant.

(B) Reverse reaction starts



At equilibrium,
rate forward = rate reverse
concentrations remain constant.

11. LE CHATELIER'S PRINCIPLE

If a change of condition is applied to a system at equilibrium, the equilibrium shifts in a direction that minimises (or opposes) the change and re-establishes a new equilibrium.

Change	Effect	Equilibrium shifts	Example
 Increase in temperature	Favours endothermic direction	Towards products (if forward reaction is endothermic)	$N_2O_4(g) \rightleftharpoons 2NO_2(g)$ (Forward is endothermic)
 Decrease in temperature	Favours exothermic direction	Towards reactants (if forward reaction is exothermic)	Same as above
 Increase in pressure (or decrease in volume)	Favours side with less moles of gas	Towards fewer moles of gaseous molecules	$N_2(g) + 3H_2(g) \rightleftharpoons 2NH_3(g)$ (Shifts towards NH_3 side)
 Decrease in pressure (or increase in volume)	Favours side with more moles of gas	Towards greater moles of gaseous molecules	Same as above
 Addition of reactant	System uses it up	Towards products	-
 Addition of product	System reduces it	Towards reactants	-

Remember!

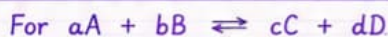
Nature always
tries to
maintain
balance. 😊

EQUILIBRIUM

At equilibrium, Gibbs free energy of the system is minimum.
 $\Delta G = 0$

12. PREDICTION OF DIRECTION OF REACTION

The reaction quotient (Q) is used to predict the direction in which a reaction will proceed to reach equilibrium.



$$Q_c = \frac{[C]^c [D]^d}{[A]^a [B]^b}$$

Comparison of Q_c with K_c	Predicted Direction
$Q_c < K_c$	Reaction proceeds forward (towards products)
$Q_c > K_c$	Reaction proceeds backward (towards reactants)
$Q_c = K_c$	Reaction is at equilibrium (no change in concentrations)

13. EQUILIBRIUM IN TERMS OF GIBBS FREE ENERGY

For a reaction at constant temperature and pressure, the change in Gibbs free energy (ΔG) is given by:

$$\Delta G = \Delta G^\circ + RT \ln Q$$

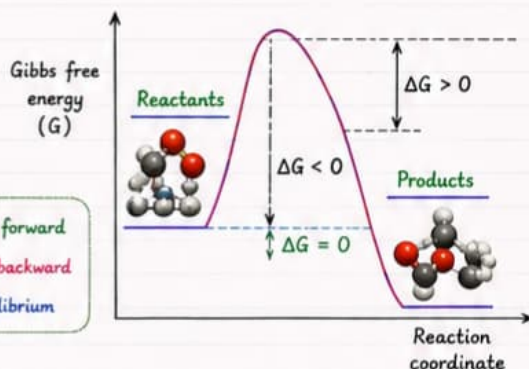
At equilibrium, $\Delta G = 0$ and $Q = K$

$$\Delta G^\circ = -RT \ln K$$

$\Delta G < 0 \rightarrow$ Reaction proceeds forward

$\Delta G > 0 \rightarrow$ Reaction proceeds backward

$\Delta G = 0 \rightarrow$ System is at equilibrium



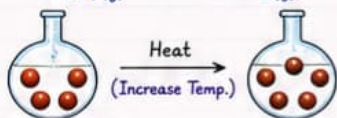
14. APPLICATION OF LE CHATELIER'S PRINCIPLE

If a system at equilibrium is subjected to a change in concentration, pressure, temperature or catalyst, the equilibrium shifts in such a direction that it counteracts the change.

Stress Applied	Change Made	Equilibrium Shifts	Example ($N_2(g) + 3H_2(g) \rightleftharpoons 2NH_3(g)$)
1. Concentration	Increase [Reactants] or Decrease [Products]	Towards products (forward direction)	Add more N_2 or H_2 More NH_3 is formed.
2. Pressure	Increase pressure (Decrease volume)	Towards side with less moles of gas	Here 4 moles (LHS) $\rightleftharpoons$ 2 moles (RHS) So, shifts towards products (RHS).
3. Temperature	Increase temperature (Add heat)	Towards endothermic direction	For this reaction, forward is exothermic. So, increase in temp. shifts equilibrium towards reactants .
4. Catalyst	Add catalyst	Does not shift equilibrium (attains it faster)	Catalyst increases the rate of both forward and reverse reactions equally. Only time to reach equilibrium changes.

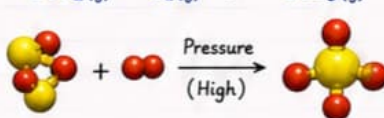
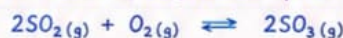
15. MORE EXAMPLES

① Decomposition of N_2O_4



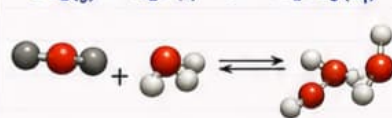
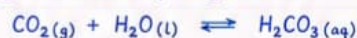
N_2O_4 (Colourless) $\rightleftharpoons$ NO_2 (Brown)
 On heating, equilibrium shifts towards NO_2 (brown colour deepens).

② Formation of SO_3 (Contact process)



High pressure shifts equilibrium towards formation of SO_3 .

③ Dissolution of CO_2 in water



Increase in CO_2 pressure shifts equilibrium towards H_2CO_3 formation.

QUICK SUMMARY

- ✓ Equilibrium is dynamic; rates of forward and reverse reactions are equal.
- ✓ Equilibrium constant (K_c) depends only on temperature.
- ✓ $K_c > 1 \rightarrow$ Products favoured, $K_c < 1 \rightarrow$ Reactants favoured.
- ✓ Q is used to predict direction of reaction.
- ✓ At equilibrium, $\Delta G = 0$ and $Q = K$.
- ✓ Le Chatelier's principle explains the effect of changes in a system.

TIPS FOR NEET

- ★ Learn K_c expression carefully.
- ★ Remember: Effect of Change $\rightarrow$ Opposes the change.
- ★ Practice more numericals on K_c and Q_c .

IMPORTANT POINTS

- Equilibrium is not the end of reaction.
- Catalyst does not change K_c .
- Only temperature affects the value of K_c .
- Nature always tries to maintain balance.

NEET EXPRESS

- Understand Equilibrium deeply, its the key to many chapters! ☆

CHAPTER - 6

EQUILIBRIUM

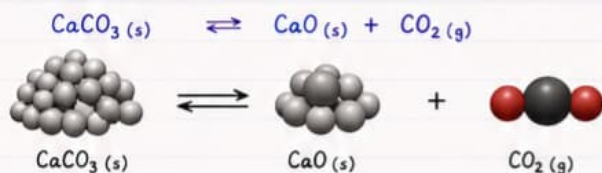
SHORT NOTES

At equilibrium macroscopic properties remain constant, but microscopic processes continue. ♥

16. HETEROGENEOUS EQUILIBRIUM - EXPRESSION

In a heterogeneous equilibrium, only the concentration of reacting species in the same phase are included in K_c .

Example :



For this reaction,

$$K_c = [\text{CO}_2]$$

★ Solids have unit activity (1) and do not appear in the equilibrium expression.

18. EQUILIBRIUM IN TERMS OF PARTIAL PRESSURE

For gaseous reaction, equilibrium constant can be expressed in terms of partial pressures (K_p).

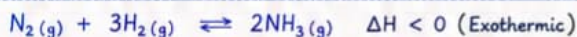


$$K_p = \frac{(P_C)^c (P_D)^d}{(P_A)^a (P_B)^b}$$

P_A, P_B, P_C, P_D are equilibrium partial pressures of the gases.

20. INDUSTRIAL APPLICATION

→ Haber process (Ammonia synthesis) :



Conditions used for maximum NH_3 yield:

- ✓ High pressure (200 - 300 atm)
- ✓ Low temperature ($\sim 450^\circ\text{C}$)
- ✓ Iron catalyst (finely divided)
- ✓ Removal of NH_3 as it forms

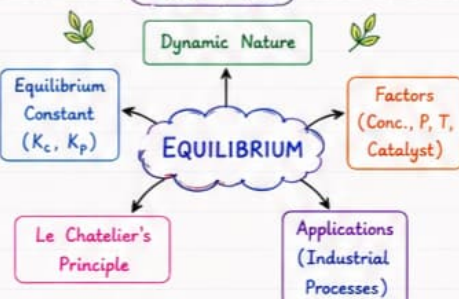


Haber Process Plant

22. IMPORTANT FACTS (ONE LINERS)

- Equilibrium is a **dynamic** state.
- At equilibrium, **forward rate = reverse rate**.
- Equilibrium constant depends only on temperature.
- Catalysts do not change K_c but help in attaining equilibrium quickly.
- Le Chatelier's principle helps predict the effect of stress on equilibrium.

MIND MAP



17. RELATION BETWEEN K_p AND K_c

For a gaseous equilibrium reaction,



$$K_p = K_c (RT)^{\Delta n_g}$$

where,

$$\Delta n_g = (c + d) - (a + b)$$

= Total moles of gaseous products
- Total moles of gaseous reactants



R = gas constant, T = absolute temperature

Special cases

- ★ If $\Delta n_g = 0$,
 $K_p = K_c$
- ★ If $\Delta n_g > 0$,
 $K_p > K_c$
- ★ If $\Delta n_g < 0$,
 $K_p < K_c$

19. EQUILIBRIUM CONSTANT AND TEMPERATURE

For most reactions, K_c increases with increase in temperature for endothermic reactions and decreases for exothermic reactions.

Type of reaction	Increase in temperature	Effect on K_c
Endothermic (Heat is absorbed)	$\uparrow T$	K_c increases
Exothermic (Heat is evolved)	$\uparrow T$	K_c decreases

REMEMBER

Temperature changes the value of K_c , but does not shift the equilibrium. ✿

21. EQUILIBRIUM VS REACTION COMPLETION

Feature	At Equilibrium	At Completion
Rate of forward reaction	Equal to rate of reverse reaction	Becomes <u>zero</u>
Rate of reverse reaction	Equal to rate of forward reaction	Becomes zero
Concentrations	Constant (non-zero)	One reactant is completely used up
Change in properties	No macroscopic change	Macroscopic change occurs

NEET TIP

- ✓ Write expressions of K_c and K_p correctly.
- ✓ Identify type of reaction (endothermic / exothermic).
- ✓ Apply Le Chatelier's principle in all types of problems.
- ✓ Practice more numericals on K_c , K_p and ΔG . ★